

DHRITISH MONDAL

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3rd Year Undergraduate

Department of Electronics and Electrical Communication Engineering

ACADEMIC QUALIFICATIONS

Year	Degree/Certificate	Institute	CPI/%
2028	B.Tech+M.Tech (Dual Degree)	Indian Institute of Technology, Kharagpur	6.33/10
2022	ISC (XII)	Calcutta Public School, Kolkata	85%
2020	ICSE (X)	Ram Mohan Mission School, Kolkata	90%

KEY PROJECTS

Production-Grade PDF RAG Chatbot (Self Project)

June 2025 – August 2025

- Constructed a modular, production-ready Document RAG architecture utilizing FastAPI, LangChain, FAISS, and Mistral-7B-Instruct to achieve intelligent semantic search and contextual QA over PDF files.
- Engineered an end-to-end processing pipeline yielding a 95% chunk retrieval success rate, handling document ingestion, embedding generation (MiniLM-L6-v2), vector indexing, and LLM synthesis.
- Leveraged advanced LangChain abstractions including RetrievalQA, ConversationalMemory, and specialized agents for document summarization, enhancing response relevance and tracking context across turns.
- Designed a responsive Streamlit interface integrated with clean API routes and metadata filtering capabilities, reducing total query latency by 30–40% through targeted pipeline optimizations.

IPL-Win Predictive App (Self Project)

May 2025 – June 2025

- Architected a predictive web platform to forecast Indian Premier League match trajectories using extensive historical datasets, utilizing feature engineering and parameter optimization to boost classifier reliability.
- Implemented a Random Forest training pipeline that assesses live match variables to calculate dynamic winning probabilities, streamlining model inference for instantaneous updates.
- Deployed an intuitive interface that delivers real-time analytical feedback based on shifting in-game metrics, optimizing back-end processing to maintain minimal application latency.

Hand Gesture Recognition Control Car (DIY Project) Prof. Sanjeev Srivastava

Jan 2024 – Mar 2024

- Designed a vision-controlled robotic vehicle using Python, deploying custom computer vision techniques to maintain high operational accuracy across diverse physical environments.
- Developed a hardware-software integration framework mapping OpenCV and machine learning outputs to physical motor control, ensuring responsive system reactions and minimized signal delay.
- Applied robust image preprocessing involving contour verification, convex hull analysis, and ambient light filtering to stabilize gesture tracking across varying distances and background noise.
- Optimized embedded power distribution to extend runtime efficiency, successfully demonstrating the prototype's viability for assistive technologies and human-robot interaction.

TECHNICAL SKILLS

Programming: C | C++ | LaTeX | MATLAB | HTML | CSS | Python | Java | Machine Learning

Frameworks: NumPy | Pandas | Scikit-learn | Flask | Matplotlib

Tools: Jupyter | Colab | VS Code | GitHub | MS Word | PowerPoint | Excel | Power BI

Software/Tools: VS Code | SQL | Jupyter Notebook

POSITIONS OF RESPONSIBILITY

Design Team Member, Spring Fest '24 (2023–24)

- Collaborated with the design team to create event props, stage elements, and promotional materials.
- Applied Photoshop and video editing skills to design visually compelling event assets.
- Assisted the sponsorship team in coordinating with partners and played an active role in securing Rs.80,000 in sponsorship funding and managed stage and ground operations during live events.

Secretary of Sports — AJCB Hall of Residence, IIT Kharagpur '25 (2024–2025)

- Led and coordinated the sports activities, managing 100+ students across Cricket, Football, Chess, and Athletics.
- Represented the hall in football in Inter-Hall General Championship, contributing to overall team performance.
- Secured 2nd place in Football Men and Bronze in Cricket Men in General Championship, IIT Kharagpur'23.

COURSE WORK

Core: Data Structures & Algorithms | Signals & Systems | Calculus | Linear Algebra & Optimization
| Analog Electronics | Digital Electronics | Probability & Statistics | Electrical Technology |
Semiconductor Devices | Control Systems | Analog Communication | Principles of Communication
| Algorithms | RF Microwave | Digital Signal Processing

EXTRA-CURRICULAR ACTIVITIES

- Participated in the Inter IIT Football '24, gaining valuable team spirit, and on-field experiences.
- Represented hall in the Inter-Hall GC Football Tournament '25, contributing actively to team strategies.
- Represented hall in the Inter-Hall GC Cricket Tournament '25, showcasing teamwork, leadership, and sportsmanship under competitive conditions.